# Land cover / land use classifier: test on 500 records

Run 20261012-151353 · 500 random records (seed 42) of 25,129 · both taxonomies, depth 3 · mistral-small-latest, 5 records/call, 8 parallel · 12 Oct 2026

**11 min 29 s**
total time (0.73 records/s)

**$0.183**
total cost ($0.37 per 1,000)

**462**
API calls

**936k**
tokens (about 1,872 per record)

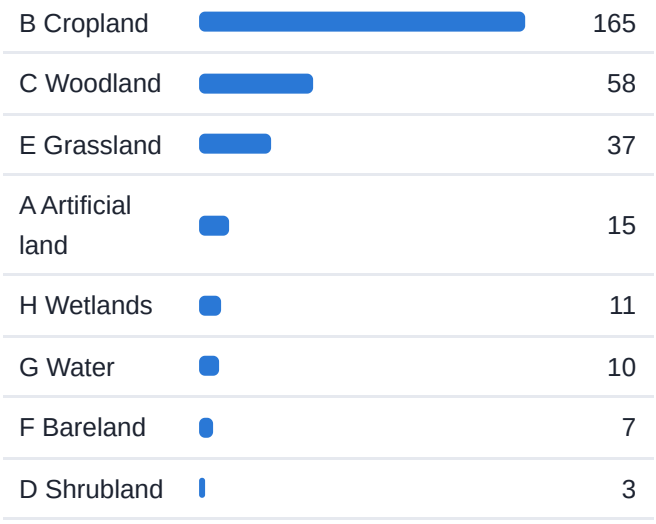
## Time and cost

| METRIC | VALUE |
|---|---|
| Wall time | 689 s (1.38 s per record) |
| Input / output tokens | 841,323 / 94,913 |
| Cost per record | $0.00037 (prices: $0.15 / $0.60 per million input / output tokens) |
| Extrapolated to all 25,129 records | about 9.6 h, about $9.20 |

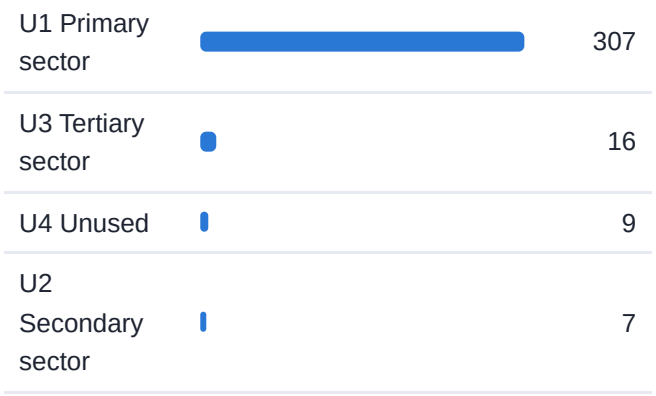
## Label coverage and confidence (default thresholds)

| | LAND COVER | LAND USE |
|---|---|---|
| Records with at least 1 label | 284 (57%) | 337 (67%) |
| Records reaching level 2 / level 3 | 37% / 20% | 52% / 42% |
| Labels per record (deepest) | 0.71 | 0.71 |
| Mean level-1 confidence | 0.94 | 0.92 |
| Level-1 labels with confidence ≥ 0.8 | 100.0% | 99.7% |

## Land cover: level 1 classes




## Land use: level 1 classes



## Accuracy 1: independent check by Claude (60 records)

The records table has no human labels. Claude labelled 60 of the 500 records (random, seed 7) at level 1 **before** seeing Mistral's answers, from the title and the first 650 characters of the abstract (Mistral saw the full text). Mistral's labels use the default thresholds. Details: `reports/claude_verification_60.csv`.

| LEVEL 1 | EXACT MATCH | PRECISION | RECALL | F1 |
|---|---|---|---|---|
| Land cover | 47/60 (78%) | 0.82 | 0.79 | **0.81** |
| Land use | 40/60 (67%) | 0.54 | 0.73 | **0.62** |

**Land cover** is good: most of the 13 differences are a second class Mistral missed (wetland, shrubland, bare soil); in 3 cases it added one Claude did not.
**Land use** has a clear problem: Mistral puts U1 (Primary sector) on soil-science records that are not about farming, forestry, fishing or mining (biochar, an E. coli lab column, soil enzymes, a global salinity map). 14 of the 20 differences are this, and U1 is 307 of 337 labelled land-use records in the whole run. **Suggested fix:** tighten the U1 hint in the Labels tab; a changed hint reclassifies the records once (about $0.18 per 500).
**Limits:** one annotator (some labels are judgment calls), level 1 only, small sample. This is a second opinion from a stronger model, not human ground truth.

## Land use: the 20 differences

| # | TITLE | CLAUDE | MISTRAL |
|---|---|---|---|
| 9 | The power of pyrolysis | - | U1 |
| 10 | Effect Of Elevated Co2 And Temperature On The Oxidative Stress Respons | - | U1 |
| 15 | International workshop from yield-based to society-based fertilizer re | U1 | - |
| 16 | Cyanobacteria for restoration of degraded soils | - | U1 |
| 20 | Physiological And Biochemical Responses Ofquercus Pubescensto Air Warm | - | U1 |
| 21 | ORCHILEAK (revision 3875) | - | U1 |
| 26 | Bacteriophages limitedly contribute to the antimicrobial resistome of | U3 | - |
| 27 | Remediation of water from perfluorooctanoic acid (PFOA) with ozonation | - | U2 |
| 29 | Nutrient Additions To A Tropical Rain Forest Drive Substantial Soil Ca | U4 | U1 |
| 37 | Biotic and abiotic factors controlling spatial variation of mean carbo | - | U1 |
| 39 | Biochar Pyrolyzed At Two Temperatures Affects Escherichia Coli Transpo | - | U1 |
| 41 | Soil scans for parcels in Netherlands (20170327_vermeulen_spieClass_3) | U1 | - |
| 42 | Different Types Of Sub-Alpine Grassland Respond Similarly To Elevated | - | U1 |
| 47 | The Invasive Shrub European Buckthorn (Rhamnus Cathartica, L.) Alters | U4 | U1 |
| 51 | Cessation of grazing causes biodiversity loss and homogenization of so | U1,U4 | U1 |
| 52 | Elevated Co2 Alleviates Damage From Potato Virus Y Infection In Tobacc | - | U1 |
| 53 | From fine to giant: multi-instrument assessment of the dust particle s | - | U1 |
| 56 | Global Soil Salinity Map | - | U1 |
| 58 | Farmwell social innovation Greece energy community | U2 | U1 |
| 59 | SUMMARY OF THE PM(T) ONLINE SURVEY RESULTS - MAY-JULY 2022 | - | U2 |

## Accuracy 2: agreement with an earlier run

Agreement with run 20261007-162217 (1 record per call) on the 100 records both runs contain. The reference is not human-verified: this measures consistency, not correctness. Precision, recall and F1 treat the reference labels as truth.

| LEVEL | LAND COVER | | | | LAND USE | | | |
|---|---|---|---|---|---|---|---|---|
| | PREC. | RECALL | F1 | EXACT | PREC. | RECALL | F1 | EXACT |
| 1 | 88% | 85% | 0.86 | 85% | 94% | 89% | 0.92 | 90% |
| 2 | 78% | 78% | 0.78 | 83% | 85% | 76% | 0.80 | 80% |
| 3 | 91% | 83% | 0.87 | 94% | 82% | 71% | 0.76 | 79% |